

Task 1: Process Synchronization and Child Process Monitoring Using waitpid()

```
root@LAPTOP-9F77QCEQ:~# nano filename.c
root@LAPTOP-9F77QCEQ:~# gcc filename.c -o filename
root@LAPTOP-9F77QCEQ:~# ./filename
[491] Parent process waiting for child 492 to finish...
Child Process: PID = 492, executing work...
Child Process: Exiting with status 0
[491] Child process 492 terminated normally with exit status: 0
root@LAPTOP-9F77QCEQ:~# |
```

Task 2: To Retrieve PATH Variable, Parse Search Directories, Locate Executables, Verify Permissions, Handle Missing Commands, Test Command Resolution

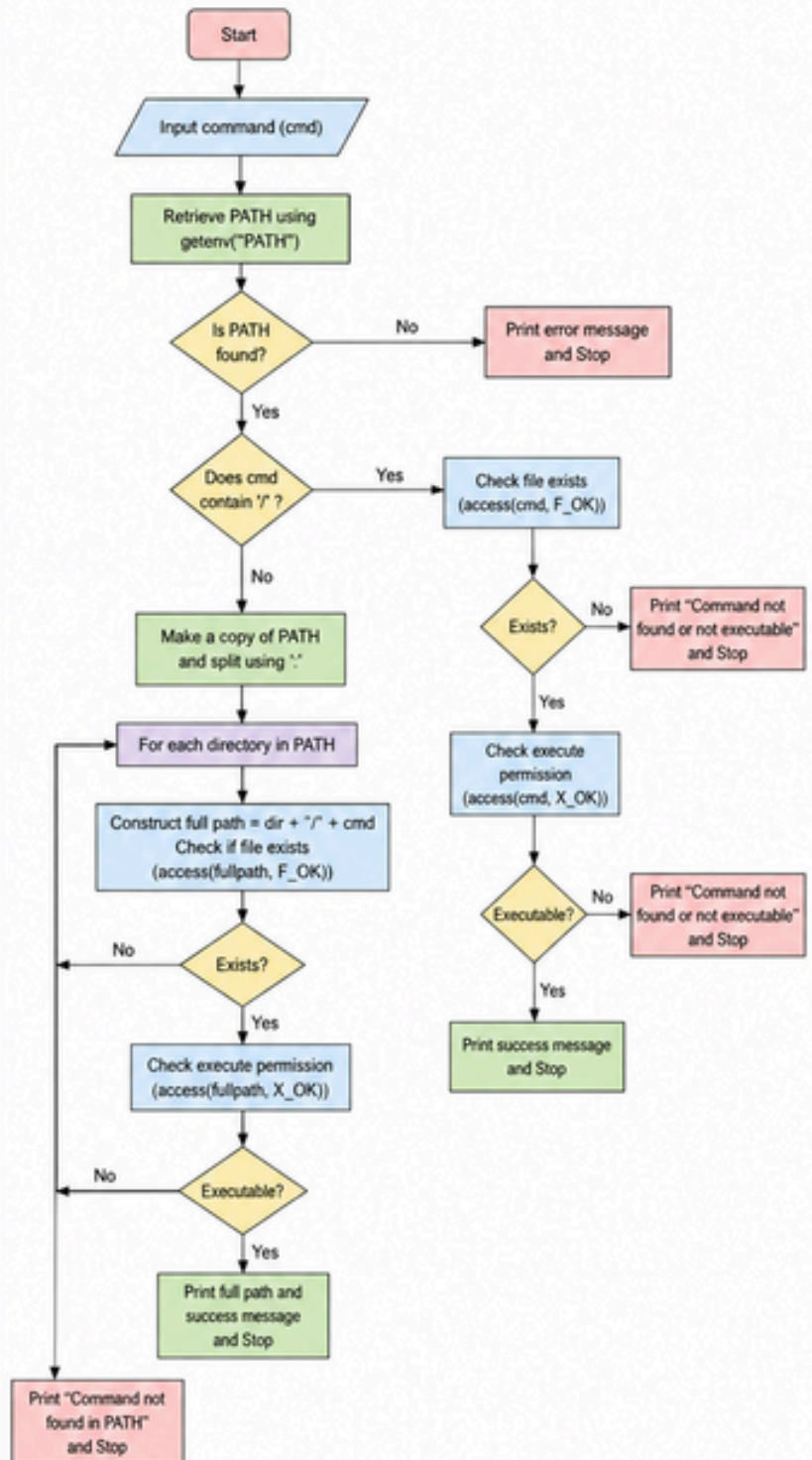
Function	Purpose
getenv()	Retrieves the PATH environment variable
strtok()	Splits PATH into individual directories
access()	Checks file existence and execute permission
snprintf()	Builds the complete executable path
strdup()	Creates a modifiable copy of PATH
free()	Releases allocated memory

Flow Chart:

ALGORITHM

1. Start
2. Input command name from user (cmd).
3. Retrieve the PATH environment variable using `getenv("PATH")`.
If PATH is not found, print error message and stop.
4. If cmd contains '/' (i.e., user gave a path):
 - 4.1. Check if the file exists using `access(cmd, F_OK)`.
 - 4.2. If it exists, check execute permission using `access(cmd, X_OK)`.
 - 4.3. If executable, print success message.
 - 4.4. Else, print "Command not found or not executable".
 - 4.5. Stop.
5. Else (cmd does not contain '/'):
 - 5.1. Make a copy of PATH.
 - 5.2. Split the PATH using ':' as delimiter to get directories.
 - 5.3. For each directory in PATH do
 - 5.3.1. Construct full path = directory + '/' + cmd.
 - 5.3.2. Check if file exists at full path using `access(fullpath, F_OK)`.
 - 5.3.3. If file exists, check execute permission using `access(fullpath, X_OK)`.
 - 5.3.4. If executable, print full path and success message. Go to Step 6.
 - 5.3.5. Else, continue with next directory.
 - 5.4. If all directories are checked and command not found, print "Command not found in PATH".
6. Stop.

FLOWCHART



Sample Output:

Case 1:

./T2 pwd

PATH = /usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin:/usr/games:/usr/local/games:/mnt/c/Program Files/WindowsApps/CanonicalGroupLimited.Ubuntu_2404.1.68.0_x64__79rhkp1fndgsc:/mnt/c/Windows/system32:/mnt/c/Windows:/mnt/c/Windows/System32/Wbem:/mnt/c/Windows/System

32/WindowsPowerShell/v1.0:/mnt/c/Windows/System32/OpenSSH:/mnt/c/Program Files/Microsoft VS
Code/bin:/mnt/c/Users/DELL/AppData/Local/Microsoft/WindowsApps:/mnt/c/Users/DELL/AppData/Local/Git
HubDesktop/bin:/snap/bin

File found: /usr/bin/pwd

Execute permission: YES

Command resolved successfully.

Case 2:

./T2 Raghu

PATH = /usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin:/usr/games:/usr/local/games:/mnt/c/Program
Files/WindowsApps/CanonicalGroupLimited.Ubuntu_2404.1.68.0_x64__79rhkp1fndgsc:/mnt/c/Windows/syste
m32:/mnt/c/Windows:/mnt/c/Windows/System32/Wbem:/mnt/c/Windows/System32/WindowsPowerShell/v1.0/
:/mnt/c/Windows/System32/OpenSSH:/mnt/c/Program Files/Microsoft VS
Code/bin:/mnt/c/Users/DELL/AppData/Local/Microsoft/WindowsApps:/mnt/c/Users/DELL/AppData/Local/Git
HubDesktop/bin:/snap/bin

Command 'Raghu' not found in PATH.

```
Parent Process
Parent PID : 668
Child PID  : 715

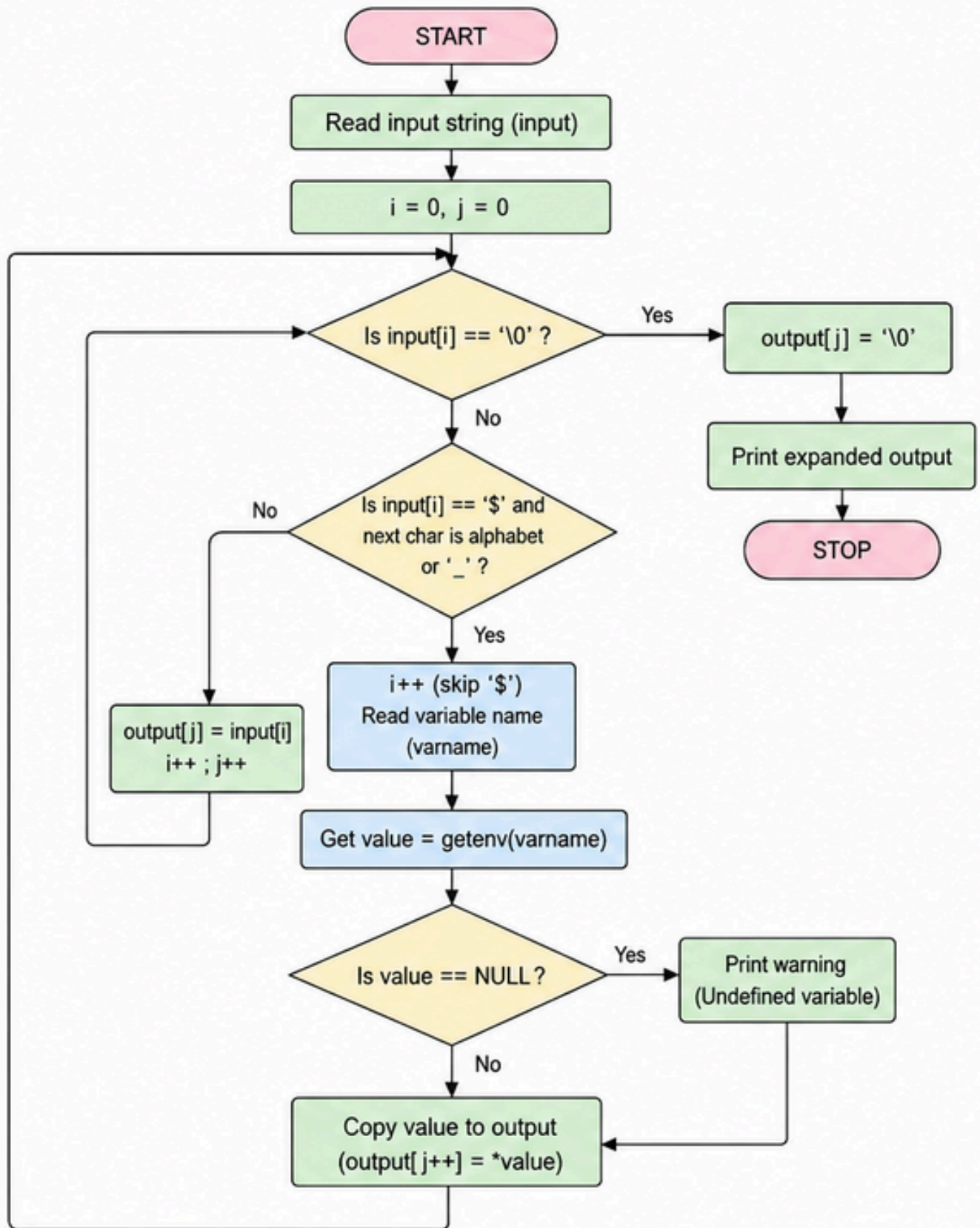
Child Process
Child PID  : 715
Parent PID : 668
/next
```

Task 3: To Detect Variable References, Expand Values, Handle Undefined Variables, Update Tokens, Support
Nested Variables, Test Expansion Logic.

Function	Purpose
getenv()	Retrieves environment-variable values
fgets()	Reads input
strcspn()	Removes newline
isalpha()	Checks alphabetic character
isalnum()	Checks alphanumeric character

Flow Chart:

FLOWCHART – 1 VARIABLE EXPANSION



Sample Output:

Case 1:

Enter a string: Raghu

Original : Raghu

Expanded : Raghu

Case 2:

Enter a string: hello \$abc

Warning: Undefined variable: abc

Original : hello \$abc

Expanded : hello

```
try: apt install <deb name>
root@LAPTOP-9F77QCEQ:~# gcc arun.c -o arun
root@LAPTOP-9F77QCEQ:~# ./arun
Enter a string: arun
Original : arun
Expanded : arun
root@LAPTOP-9F77QCEQ:~#
```

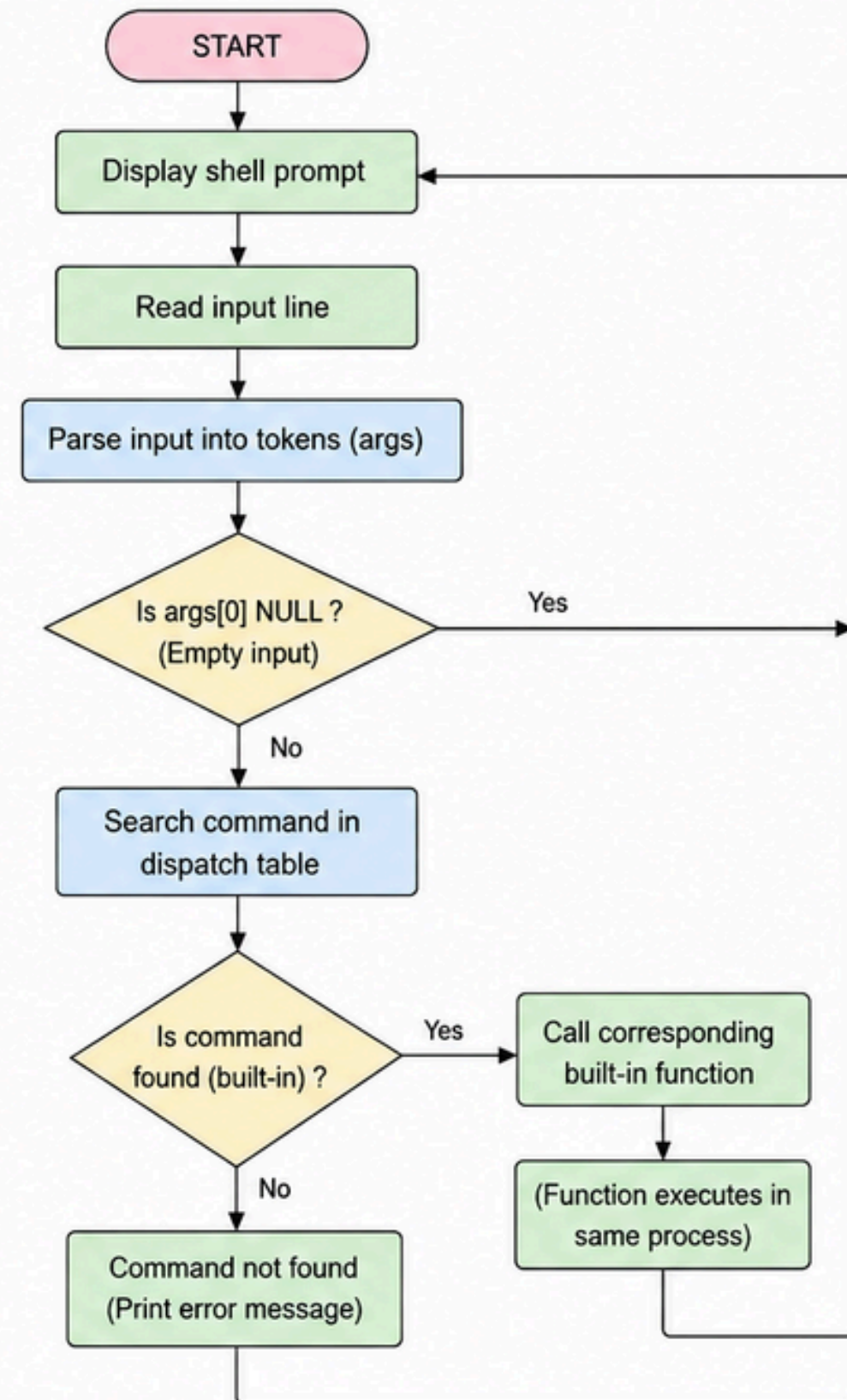
Task 4: To Identify Built-ins, Create Dispatch Table, Execute In-Process Built-ins, Handle Invalid Commands, Maintain State, Test Dispatch Logic

Function	Purpose
chdir()	Changes the current working directory
getcwd()	Gets the current working directory
getenv()	Retrieves environment variables
exit()	Terminates the process

strcmp()	Compares command names
strtok()	Separates command into tokens

Flow Chart:

FLOWCHART – 2 BUILT-IN COMMAND DISPATCH



Sample output:

```
myshell> pwd
```

```
/home/student
```

```
myshell> echo Hello World
```

```
Hello World
```

```
myshell> cd /tmp
```

```
myshell> pwd
```

```
/tmp
```

```
myshell> abc
```

```
Command not found: abc
```

```
myshell> exit
```

```
Exiting shell...
```

```
myshell> pwd
/root
myshell> echo Hello word
Hello word
myshell> pwd
/root
myshell> abc
Command not found: abc
myshell> exit
Exiting shell...
root@LAPTOP-9F77QCEQ:~# |
```